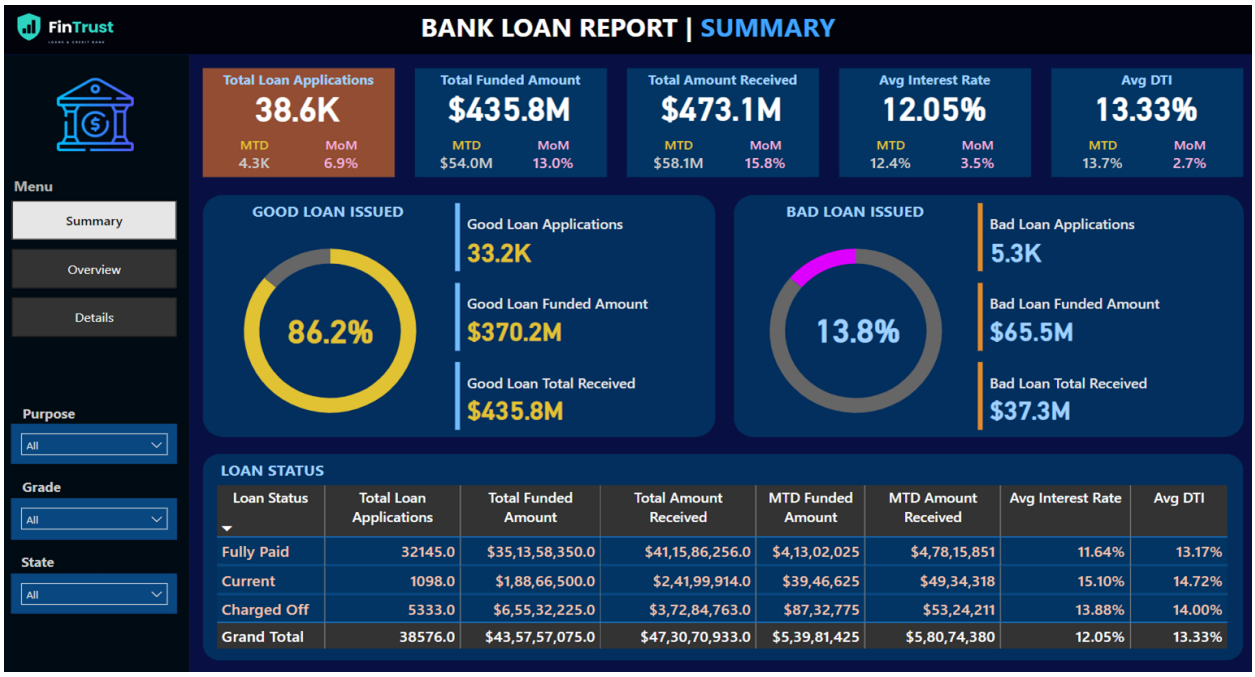
BANK LOAN ANALYSIS

Full Project Explanation & Interview Preparation

End-to-End Data Analytics Project | Finance / Banking Domain



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1. PROJECT OVERVIEW

Bank Loan Analysis is an end-to-end analytics project designed to monitor lending activity, portfolio health, borrower characteristics and loan performance. The project starts with raw loan data, performs analysis in Python, uses MS SQL Server for business queries and validation, and presents the results through interactive Power BI dashboards.

The documented dataset contains 38,576 rows and 24 columns. Important fields include loan status, state, employment length, grade, home ownership, issue date, purpose, term, annual income, DTI, interest rate, loan amount and total payment.

Business problem: The business needs a comprehensive report to track key loan KPIs, understand Good Loan versus Bad Loan performance, identify trends and compare lending activity across different borrower and loan dimensions.

Business objective: Turn raw loan records into reliable metrics and interactive reporting that can support portfolio monitoring, risk assessment, lending performance analysis and data-driven decision-making.

2. END-TO-END PROJECT WORKFLOW

Step 1 — Understand the Business Requirement
Start by identifying what the business wants to measure: applications, funded amount, amount received, interest rate, DTI, Good/Bad Loans and trend dimensions.

Step 2 — Explore the Data in Python
Load the dataset, inspect shape, columns, data types and descriptive statistics, then understand the meaning and quality of important fields.

Step 3 — Prepare and Analyze the Data
Check dates, numeric fields, categories and business definitions. Perform the required calculations and validate outputs.

Step 4 — Load into MS SQL Server
Create the database/table and load the loan data so business questions can be answered with SQL.

Step 5 — Write SQL Queries
Use COUNT, SUM, AVG, WHERE, GROUP BY, ORDER BY and date functions to calculate the required metrics and dimensional analysis.

Step 6 — Validate Results
Compare SQL results with Python calculations and Power BI outputs. The project documentation specifically describes comparing SQL results with BI tools.

Step 7 — Build Power BI Model and Measures
Connect Power BI to SQL Server, use Power Query where needed, create date/time logic and DAX measures, then build KPI cards and charts.

Step 8 — Build Three Dashboards
Summary for executive KPIs, Overview for trends/segments and Details for record-level analysis.

3. KPI FOUNDATION — WHAT EACH METRIC MEANS

1. Total Loan Applications

Meaning: The total number of loan records/applications in the selected period or filter context.

Formula: **COUNT(id)**

Project example: The dashboard shows approximately **38.6K** applications; the Python analysis contains exactly **38,576** records.

Simple example: If a bank receives 1,000 applications, Total Loan Applications = 1,000.

2. Month-to-Date (MTD) Loan Applications

Meaning: Number of loan applications in the current month being analyzed.

SQL approach used in the project: COUNT(id) filtered by the relevant issue-date month.

Project example: December has **4,314** applications. November has **4,035**.

MTD is useful because it tells management how the current month's lending volume is progressing.

3. Month-over-Month (MoM) Change

Meaning: Percentage change between the current month and previous month.

Formula: **(Current Month – Previous Month) / Previous Month × 100**

Project example: (4,314 – 4,035) / 4,035 × 100 ≈ **6.9%**, matching the dashboard's MoM application change.

4. FINANCIAL KPIs — FUNDED, RECEIVED & INTEREST

4. Total Funded Amount

Meaning: Total principal amount disbursed/funded through the loans.

Formula: **SUM(loan_amount)**

Project example: Total Funded Amount is approximately **$435.8M**. The SQL result is about 435.76M in the source data.

Simple example: If 3 loans of $10,000, $15,000 and $20,000 are funded, total funded amount = $45,000.

5. MTD Funded Amount

Meaning: Total loan amount funded during the current month.

Project example: December MTD funded amount is approximately **$54.0M**.

6. Total Amount Received

Meaning: Total payment/collection received from borrowers.

Formula: **SUM(total_payment)**

Project example: Total Amount Received is approximately **$473.1M**.

Simple example: If three borrowers have paid $8,000, $12,000 and $15,000, total amount received = $35,000.

7. MTD Amount Received

Meaning: Borrower payments received during the current month.

Project example: December MTD amount received is approximately **$58.1M**.

8. Average Interest Rate

Meaning: Average interest rate across the loan portfolio.

Formula: **AVG(int_rate) × 100** when int_rate is stored as a decimal.

Project example: Average Interest Rate = **12.05%**; the Python descriptive statistics show a mean int_rate of approximately 0.120488.

Simple example: For rates 10%, 12% and 14%, average interest rate = (10+12+14)/3 = 12%.

5. RISK & PORTFOLIO QUALITY METRICS

9. Average Debt-to-Income Ratio (DTI)

Meaning: Average borrower debt burden relative to income. In the domain documentation, DTI is described as monthly debt payments relative to monthly income; lower DTI generally indicates better repayment capacity.

Formula used in the project: **AVG(dti) × 100**.

Project example: Average DTI = **13.33%**. The Python descriptive statistics show mean dti ≈ 0.133274.

Simple example: If three borrowers have DTI values of 10%, 15% and 20%, average DTI = 15%.

10. Good Loan Application Percentage

Definition in the project: Good Loans are loans with status **Fully Paid** or **Current**.

Formula: **Good Loan Applications / Total Applications × 100**.

Project example: 33,243 Good Loan applications / 38,576 total applications × 100 = **86.18%**.

11. Good Loan Applications

Counts applications where loan_status is Fully Paid or Current. Project result: **33,243**.

12. Good Loan Funded Amount

Sum of loan_amount for Fully Paid and Current loans. Project result: approximately **$370.22M**.

13. Good Loan Total Received Amount

Sum of total_payment for Fully Paid and Current loans. Project result: approximately **$435.79M**.

6. BAD LOAN METRICS

14. Bad Loan Application Percentage

Definition in the project: Bad Loans are loans with status **Charged Off**.

Formula: **Bad Loan Applications / Total Applications × 100**.

Project example: 5,333 / 38,576 × 100 = **13.82%**.

15. Bad Loan Applications

Counts applications where loan_status = Charged Off. Project result: **5,333**.

16. Bad Loan Funded Amount

Sum of loan_amount for Charged Off loans. Project result: approximately **$65.53M**.

17. Bad Loan Total Received Amount

Sum of total_payment for Charged Off loans. Project result: approximately **$37.28M**.

How to interpret this section in an interview: I used loan status to separate the portfolio into Good and Bad Loans. This gives management a quick view of portfolio quality and the amount of capital associated with loans that are currently performing versus charged off.

Important: The project uses these categories as reporting definitions. They should not be presented as a complete credit-risk model.

7. LOAN STATUS GRID — HOW TO EXPLAIN IT

The Summary dashboard contains a Loan Status grid. The project requirements specify metrics including Total Loan Applications, Total Funded Amount, Total Amount Received, MTD Funded Amount, MTD Amount Received, Average Interest Rate and Average DTI by loan status.

Fully Paid: 32,145 applications; funded amount approximately $351.36M; amount received approximately $411.59M; average interest rate approximately 11.64%; average DTI approximately 13.17%.

Current: 1,098 applications; funded amount approximately $18.87M; amount received approximately $24.20M; average interest rate approximately 15.10%; average DTI approximately 14.72%.

Charged Off: 5,333 applications; funded amount approximately $65.53M; amount received approximately $37.28M; average interest rate approximately 13.88%; average DTI approximately 14.00%.

Example interpretation: If Charged Off loans show a lower amount received relative to funded amount, that indicates weaker collection performance for that status and is a useful portfolio-monitoring signal.

The SQL query groups by loan_status and calculates COUNT(id), SUM(total_payment), SUM(loan_amount), AVG(int_rate × 100) and AVG(dti × 100).

8. OVERVIEW DASHBOARD — EACH CHART EXPLAINED

18. Monthly Trends by Issue Date

Purpose: Identify seasonality and long-term trends in lending activity.

Project example: Applications increase from **2,332 in January** to **4,314 in December**. This shows a strong upward pattern in application volume over the year.

19. Regional Analysis by State

Purpose: Identify states with significant lending activity and regional differences. SQL groups by address_state and calculates applications, funded amount and amount received.

20. Loan Term Analysis

Purpose: Understand the distribution of loans by repayment term. The project has 36-month and 60-month categories.

Project example: 36-month loans = **28,237** applications; 60-month loans = **10,339** applications.

21. Employee Length Analysis

Purpose: Understand lending volume across borrower employment histories.

Project example: The 10+ years category has **8,870** applications in the SQL output.

22. Loan Purpose Analysis

Purpose: Understand why borrowers seek financing and compare applications, funded amount and received amount by purpose.

Project example: Debt consolidation is the largest purpose in the dashboard, with roughly **18K** applications.

9. HOME OWNERSHIP & DETAILS DASHBOARD

23. Home Ownership Analysis

Purpose: Understand lending activity by borrower home-ownership category and assess how applications and funding differ across groups.

The SQL analysis groups by home_ownership and calculates Total Loan Applications, Total Funded Amount and Total Amount Received.

Project example: The SQL output shows **RENT** as the largest category by applications, with **18,439** applications.

24. Details Dashboard

The Details dashboard is the record-level view. It provides fields such as loan ID, purpose, home ownership, grade, sub-grade, issue date, funded amount, interest rate, installment and amount collection.

Why it matters: Summary dashboards answer 'What is happening?' Overview answers 'Where and across which segments is it happening?' Details helps answer 'Which records are contributing to the result?'

The project requirements describe the Details dashboard as a consolidated, user-friendly interface for accessing essential loan information and supporting detailed portfolio and borrower analysis.

10. SQL ANALYSIS — HOW TO EXPLAIN YOUR QUERIES

Total Applications

```
SELECT COUNT(id) AS Total_Applications FROM bank_loan_data;
```

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount FROM bank_loan_data;
```

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected FROM bank_loan_data;
```

Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications FROM bank_loan_data WHERE loan_status IN
('Fully Paid','Current');
```

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications FROM bank_loan_data WHERE loan_status =
'Charged Off';
```

Dimension analysis pattern

```
SELECT term, COUNT(id), SUM(loan_amount), SUM(total_payment) FROM bank_loan_data
GROUP BY term ORDER BY term;
```

The same pattern is used for state, employee length, purpose and home ownership. Filters can also be applied—for example, the project documentation demonstrates filtering by Grade A and grouping results by purpose.

11. POWER BI — HOW THE DASHBOARD WAS BUILT

Data Connection: Connect Power BI to the MS SQL Server data source.

Power Query: Apply required transformations and preparation steps.

Data Model: Build a usable analytical model and date logic where required.

DAX Measures: Create calculations for KPIs, totals, percentages and time-intelligence requirements.

Visuals: KPI cards, donut charts, line charts, filled map, bar charts, treemap and table/grid views.

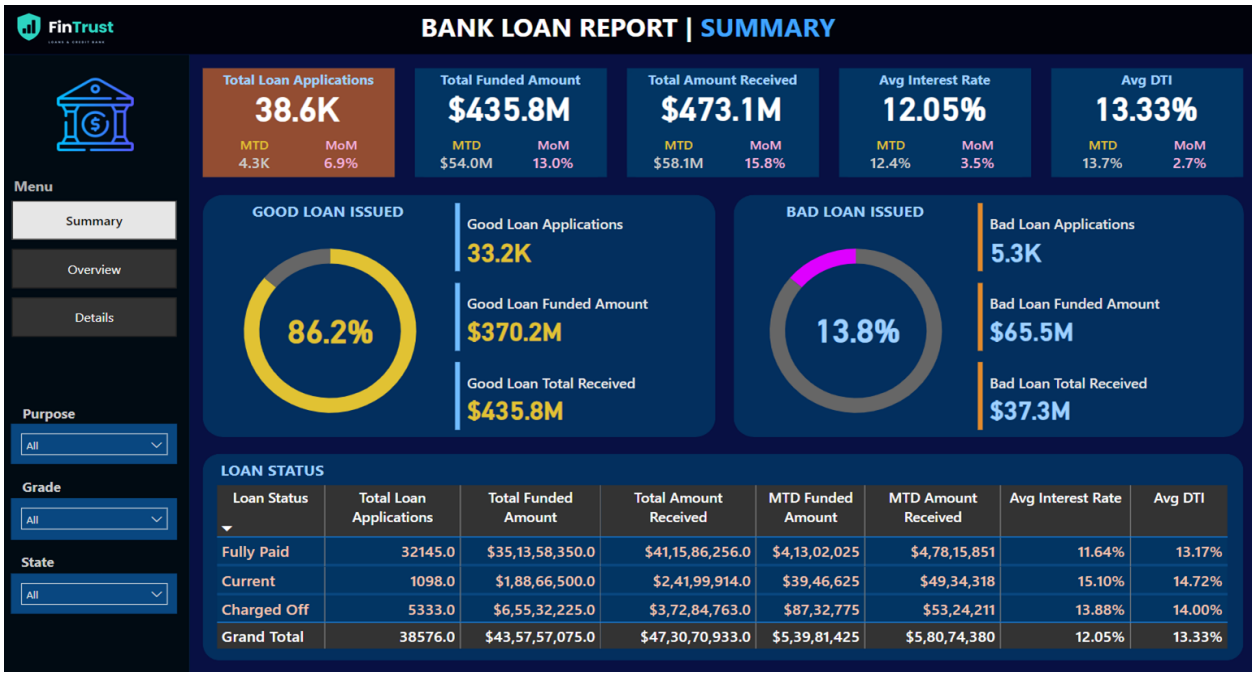
Navigation: Provide Summary, Overview and Details pages so users can move from executive KPIs to analytical and record-level views.

Filters: The dashboards include filters such as Purpose, Grade, State and Good/Bad Loan. The project documentation also describes validating filtered results by modifying SQL queries.

Validation approach: Pick one KPI in Power BI, reproduce it with SQL, then compare. For example, Good Loan Applications should equal the SQL count where loan_status is Fully Paid or Current.

12. DASHBOARD WALKTHROUGH

The three dashboards form a logical story: Summary → Overview → Details.



Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Funded Amount	MTD Amount Received	Avg Interest Rate	Avg DTI
Fully Paid	32145.0	$35,13,58,350.0	$41,15,86,256.0	$4,13,02,025	$4,78,15,851	11.64%	13.17%
Current	1098.0	$1,88,66,500.0	$2,41,99,914.0	$39,46,625	$49,34,318	15.10%	14.72%
Charged Off	5333.0	$6,55,32,225.0	$3,72,84,763.0	$87,32,775	$53,24,211	13.88%	14.00%
Grand Total	38576.0	$43,57,57,075.0	$47,30,70,933.0	$5,39,81,425	$5,80,74,380	12.05%	13.33%

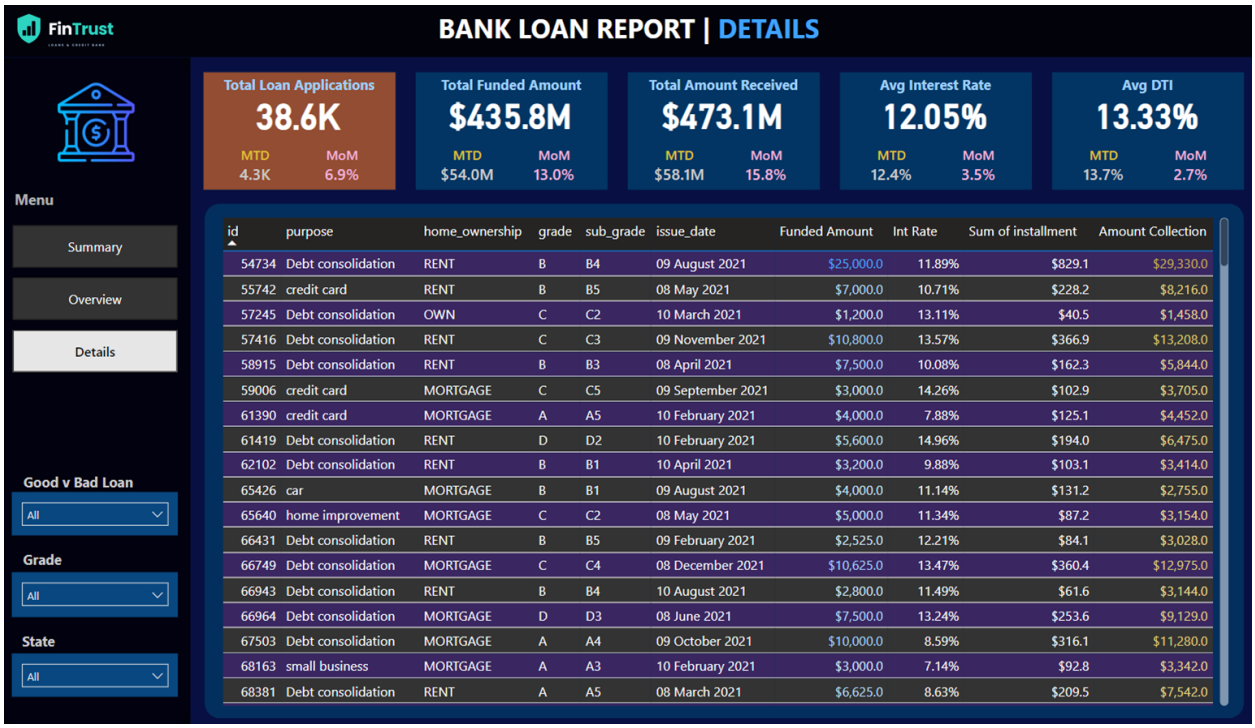
Summary: Executive KPI view, Good vs Bad Loans and Loan Status grid.

12. DASHBOARD WALKTHROUGH — OVERVIEW



Overview: Monthly trends, state analysis, term, employee length, purpose and home ownership.

12. DASHBOARD WALKTHROUGH — DETAILS



id	purpose	home_ownership	grade	sub_grade	issue_date	Funded Amount	Int Rate	Sum of installment	Amount Collection
54734	Debt consolidation	RENT	B	B4	09 August 2021	$25,000.0	11.89%	$829.1	$29,330.0
55742	credit card	RENT	B	B5	08 May 2021	$7,000.0	10.71%	$228.2	$8,216.0
57245	Debt consolidation	OWN	C	C2	10 March 2021	$1,200.0	13.11%	$40.5	$1,458.0
57416	Debt consolidation	RENT	C	C3	09 November 2021	$10,800.0	13.57%	$366.9	$13,208.0
58915	Debt consolidation	RENT	B	B3	08 April 2021	$7,500.0	10.08%	$162.3	$5,844.0
59006	credit card	MORTGAGE	C	C5	09 September 2021	$3,000.0	14.26%	$102.9	$3,705.0
61390	credit card	MORTGAGE	A	A5	10 February 2021	$4,000.0	7.88%	$125.1	$4,452.0
61419	Debt consolidation	RENT	D	D2	10 February 2021	$5,600.0	14.96%	$194.0	$6,475.0
62102	Debt consolidation	RENT	B	B1	10 April 2021	$3,200.0	9.88%	$103.1	$3,414.0
65426	car	MORTGAGE	B	B1	09 August 2021	$4,000.0	11.14%	$131.2	$2,755.0
65640	home improvement	MORTGAGE	C	C2	08 May 2021	$5,000.0	11.34%	$87.2	$3,154.0
66431	Debt consolidation	RENT	B	B5	09 February 2021	$2,525.0	12.21%	$84.1	$3,028.0
66749	Debt consolidation	MORTGAGE	C	C4	08 December 2021	$10,625.0	13.47%	$360.4	$12,975.0
66943	Debt consolidation	RENT	B	B4	10 August 2021	$2,800.0	11.49%	$61.6	$3,144.0
66964	Debt consolidation	MORTGAGE	D	D3	08 June 2021	$7,500.0	13.24%	$253.6	$9,129.0
67503	Debt consolidation	MORTGAGE	A	A4	09 October 2021	$10,000.0	8.59%	$316.1	$11,280.0
68163	small business	MORTGAGE	A	A3	10 February 2021	$3,000.0	7.14%	$92.8	$3,342.0
68381	Debt consolidation	RENT	A	A5	08 March 2021	$6,625.0	8.63%	$209.5	$7,542.0

Details: Record-level loan attributes for drill-down and detailed investigation.

13. INTERVIEW EXPLANATION — STAR METHOD

S — Situation

“I worked on an end-to-end Bank Loan Analysis project in the finance domain. The business needed visibility into loan applications, funding, collections, portfolio quality and lending trends.”

T — Task

“My task was to analyze the loan dataset, calculate the required KPIs, separate Good and Bad Loans, answer business questions with SQL and build an interactive Power BI reporting solution.”

A — Action

“I first explored the 38,576-row dataset in Python and reviewed its 24 columns and data types. I then loaded the data into MS SQL Server and wrote queries using COUNT, SUM, AVG, filters, GROUP BY and date functions. I validated the calculations and connected SQL Server to Power BI. In Power BI, I created KPI measures and built three pages: Summary, Overview and Details, with filters and navigation.”

R — Result

“The final dashboard provided a single interactive view of 38.6K applications, about $435.8M funded, about $473.1M received, a 12.05% average interest rate and a 13.33% average DTI. It also showed 86.18% Good Loans versus 13.82% Bad Loans and allowed users to analyze trends by month, state, term, employee length, purpose and home ownership.”

14. COMMON INTERVIEW QUESTIONS & STRONG ANSWERS

Q1. Why did you use SQL if you already had Python?

“Python helped me explore and understand the dataset, while SQL gave me a structured way to answer business questions directly from the database. I also used SQL as a validation layer for the Power BI metrics.”

Q2. How did you validate your dashboard?

“I reproduced important KPIs using SQL and compared them with Power BI. For example, I validated Good Loan Applications by counting records with Fully Paid or Current status.”

Q3. What is the difference between funded amount and amount received?

“Funded amount is the loan principal disbursed, while amount received represents payments collected from borrowers. Comparing them helps understand collection and repayment performance.”

Q4. Why are Good Loans Fully Paid and Current?

“That was the business definition provided for this project. I used Fully Paid and Current as the Good Loan category and Charged Off as Bad Loan.”

Q5. What was your most important insight?

“The portfolio had a high Good Loan share of 86.18%, while 13.82% of applications were Charged Off. I would use that as a starting point for deeper analysis of the drivers of charged-off loans by grade, DTI, purpose, term and borrower characteristics.”

15. 60-SECOND PROJECT PITCH

“This was an end-to-end Bank Loan Analysis project in the finance domain. I started by understanding the business requirements and exploring a 38,576-row, 24-column dataset in Python. I analyzed the data and then loaded it into MS SQL Server, where I wrote queries for core KPIs such as loan applications, funded amount, amount received, average interest rate, DTI and Good versus Bad Loans. I validated the results and connected SQL Server to Power BI. I built three interactive dashboards: Summary for executive KPIs and portfolio quality, Overview for monthly, geographic and borrower-segment analysis, and Details for record-level investigation. The final dashboard showed about 38.6K applications, $435.8M funded, $473.1M received, 12.05% average interest and 13.33% average DTI, with 86.18% Good Loans and 13.82% Bad Loans. The project helped me demonstrate the complete workflow from business requirement to data analysis, SQL, validation and BI storytelling.”

Final interview rule

Do not only describe charts. Always connect a metric to **business meaning → calculation → validation → insight → decision**.

For example: “The Good Loan percentage is 86.18%. I calculated it as Fully Paid + Current applications divided by total applications. I validated the count in SQL, displayed it in Power BI and used it to monitor overall portfolio quality.”

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